

Baani Kaur Pasrija

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EDUCATION

NYU Tandon School of Engineering

Master of Science in Computer Science

Brooklyn, NY

May 2027

Courses: Software Engineering, Artificial Intelligence, Machine Learning

New York University

Bachelor of Arts in Computer Science and Economics | GPA: 3.5

New York, NY

May 2024

Other Relevant Courses: Data Structures, Algorithms, Computer Systems Organization, Operating Systems, Computer Graphics, Discrete Mathematics, Computer Security, Linear Algebra, Calculus I-III.

SKILLS

Technical Skills: *Languages:* Python, JavaScript, TypeScript, Java, C/C++, HTML/CSS, R, BASH, LaTeX | *Frameworks / Libraries:*

React, React Native, Next.js, Flask, Django, PyTorch, Selenium, Jinja2, NumPy, Pandas | *Databases:* SQL, SQLite, MongoDB

DevOps & Cloud: Git / GitHub, GitHub Actions (CI/CD), Docker, AWS S3, DigitalOcean, REST APIs

Tools: Jira, Confluence, Figma

WORK EXPERIENCE

Backend Developer (Python) — Insurance Project

February 2025 – June 2025

Virim Infotech

Indore, India

- Developed and maintained backend services in Python, including API design, database integration, and efficient data handling for insurance workflows.
- Applied CI/CD practices (GitHub Actions, automated builds/tests) to streamline deployment pipelines and ensure reliable delivery of new features.
- Participated in code reviews, improving code quality, maintainability, and team collaboration.
- Implemented scalable backend components to support increasing data volume, performance requirements, and web application growth.

Full Stack Software Developer

September 2024 – December 2024

Autism Technology Research Syndicate (AUTRS)

Chesterbrook, PA

- Collaborated on the SEAL project, developing features for a teaching app aimed at improving curriculum accessibility for educators.
- Implemented the interface for teachers to input curriculum, integrating it with existing backend systems to ensure seamless database updates.
- Developed a backend page direction function to dynamically route users to the appropriate page based on input/output IDs, ensuring accurate display of associated curriculum content.

Robotic Process Automation Intern

Jun. 2023 – Aug. 2023

Virim Infotech

Indore, India

- Developed a strong foundation in robotic process automation.
- Increased business operation efficiency by 80% through participation in the creation and deployment of unattended bots hosted on servers to meet the unique needs of our clients.
- Supported data extraction and invoice processing by working with Robotic Enterprise Framework, attended and unattended bots, time and event triggers, dispatcher and performer bots, and web/screen scraping.

PROJECTS

TikTalk | *Flask, JavaScript, Docker, Continuous Integration / Continuous Deployment, Digital Ocean*

- Developed backend logic to generate varied AI conversational personas using LangChain.
- Built a multi-user chat interface and integrated CI/CD pipelines using GitHub Actions and implemented unit testing.

Sing & Sync | *Flask, Docker, AWS S3, Unit Testing, CI / CD*

- Implemented backend features for note onset detection, tempo estimation, and music signal processing.
- Managed CRUD operations, S3 media storage, and wrote unit tests to validate ML powered audio-to-MIDI conversion.

ArtRoam | *Flask, AWS S3, MongoDB, HTML, Jinja2, CSS*

- Built a social platform for artists with artwork uploads (camera/file), posts, and user galleries.
- Developed profile customization features and integrated SHA-1 encrypted user authentication.

CaptionNet | *PyTorch, Neural Networks (CNNs & RNNs), Natural Language Processing (NLP)*

- Built an image caption generator using CNN feature extraction and RNN/LSTM caption modeling.
- Designed data preprocessing, trained the model on MNIST variants, and evaluated caption accuracy and performance.